

Baptiste PRAS

Master's student in Artificial Intelligence

✉ baptiste.pras@universite-paris-saclay.fr

🔗 baptistepras.fr/

🌐 <https://github.com/baptistepras>

🌐 <https://www.linkedin.com/in/baptiste-pras/>

EDUCATION

Master's degree in Artificial Intelligence (PhD Track) <i>University Paris-Saclay</i>	Sep 2025 – Present Orsay, France
Magistère in Computer Science (Honors Research Program) <i>University Paris-Saclay</i> Graduated with Honors	Sep 2024 – Aug 2025 Orsay, France
Double Bachelor's Degree in Computer Science and Mathematics <i>University Paris-Saclay</i>	Sep 2022 – Aug 2024 Orsay, France
Intensive English Language Program <i>EF New York</i> Achieved C2 Proficiency	Sep 2019 – May 2020 New York, USA

PUBLICATIONS

- Point-Based Counting of Cereals and Legumes in Intercropped Fields**
Junior Conference on Data Science & Engineering (Poster), 2026
- Fine-Grained Mention-Level Analysis of Biomedical Entity Linking Models**
Medical Informatics Europe (Full Paper), 2026
- Revisiting Class Imbalance Ratios in Imbalanced Learning**
Junior Conference on Data Science & Engineering (Full Paper), 2025

PROFESSIONAL EXPERIENCE

- Research Internship** May 2026 – Aug 2026
Inria TAU Team - LISN Orsay, France
- Designed an automated per-species counting pipeline for wheat and pea in intercropped plots from UAV and smartphone imagery. Benchmarked few-shot and crowd-counting baselines with custom preprocessing: soil filtering, isotropic tiling, and leaf-tip point annotations. Worked with a finetuned point-query transformer and evaluated optimal scaling and edge filtering, reaching 4.69% MAPE with 74.2% F1 on wheat and 7.83% MAPE with 86.2% F1 on pea; published a paper in JDSE 2026.
- Contact: François Landes (LISN, Université Paris-Saclay)
- Supervised Research Project** Jan 2026 – Mar 2026
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Investigated Biomedical Entity Linking (BEL) strategies to normalize PICO entities (Population, Intervention, Comparison, Outcome) to the MeSH knowledge base. Developed and evaluated a hybrid normalization pipeline, comparing a custom rule-based script with a state-of-the-art graph-based entity linking model.
- Research Internship** May 2025 – Aug 2025
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Analyzed biomedical entity linking models on the BELB benchmark, focusing on generalization to rare or complex mentions. Developed quantitative and visual analyses of dataset characteristics (mention length, ambiguity, frequency) and their impact on prediction quality. Compared recent models, identified consistent weaknesses, and proposed improvements; published a paper in MIE 2026.
- Contact: Nona Naderi (LISN, Université Paris-Saclay)
- Supervised Research Project** Jan 2025 – May 2025
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Studied the impact of class imbalance on classification tasks using a spherical Teacher-Student perceptron. Conducted experiments in Python (Scikit-Learn, NumPy, Matplotlib) with different noise levels, loss functions, and training methods (GD, Langevin dynamics). Demonstrated that the optimal imbalance ratio in training sets differs from 0.5 in class-imbalanced problems; published a paper in JDSE 2025.

- Evaluated and refined LLM reasoning trajectories across complex coding and mathematical tasks under rigorous alignment protocols; designed adversarial prompts and assessed multi-step outputs to mitigate hallucinations and improve factual grounding for RLHF pipelines.

PROJECTS

Structure Detection in Fusion Plasma Simulations 🔗

- Ranked 4th out of 94 competitors (81% AP50) in detecting blob structures from noisy fusion plasma frames under an extreme 30-image regime. Designed an iterative pseudo-labeling pipeline pairing a YOLOv8 backbone with a custom MLP post-filter and Sobel gradient descriptors to suppress background artifacts.

Measuring Market Impact of Financial News 🔗

- Engineered a computationally frugal NLP framework converting raw filings into structured (date, ticker, impact) tuples; implemented hierarchical Map-Reduce summarization with compact Transformers, auditing numeric fidelity via an LLM-as-a-judge suite and modeling next-day abnormal returns (0.59 ROC-AUC).

Scientific Citation Information Retrieval & Reranking 🔗

- Built a multi-stage citation retrieval pipeline on 20,000 papers, improving ranking from 0.45 to 0.67 NDCG@10 (+49%). Fused sparse lexical scores (BM25, citation contexts) with dense embeddings (BGE-large, SciNCL) and metadata into a non-linear XGBRanker over 16 features.

Unpaired Image-to-Image Translation (CycleGAN from Scratch) 🔗

- Re-implemented CycleGAN in pure NumPy without autograd or PyTorch. Programmed manual analytical backward passes (im2col convolutions, instance norm, Adam) for ResNet generators and PatchGAN discriminators; validated cycle consistency ($L_1 \approx 0.20$) on horse2zebra and apple2orange.

Selected Machine Learning Projects 🔗

- Engineered end-to-end predictive pipelines: Traffic Sign Recognition (>95% accuracy using custom feature extraction) and NBA MVP prediction (>80% accuracy on historical stats via multi-model Scikit-Learn benchmarks).

SKILLS

Programming:

- Python (PyTorch, Scikit-Learn, NumPy), Bash, C/C++, Java, OCaml, SQL

Machine Learning & Vision:

- Deep Learning, Natural Language Processing, Computer Vision, Signal Processing

Tools:

- Linux, Slurm (Cluster Computing), Git, Hugging Face, OpenCV, L^AT_EX

ACHIEVEMENTS

Winner of an AI competition by designing the best-performing Dual Sudoku agent

3x Prologon finalist & 1x Gradient finalist, national algorithmic programming contests

LANGUAGES

French

Native

Russian

Conversational

English

Bilingual Proficiency

TOEFL iBT: 108/120

TOEIC: 990/990